BSNS8051- Supply Chain Design

Assignment 1 **Semester 1, 2025**

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Due date and time:	**Oral presentation in Week 5, and a soft copy of the report submitted to Turnitin through the course Moodle page by 30th March. 2025.**	
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Expected number of hours to spend on this assessment:	**40**	
Word limit:	4000 words	
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Lecturer:	**V.R Sanjeev Kumar**	
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Genesis Energy: Supply Chain Analysis and Recommendations

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1. Introduction

Genesis Energy Limited is a leading energy provider in New Zealand. Its work covers electricity production, retail sales, and gas exploration (Genesis Energy, 2023). Genesis Energy started in the year 1998. Business is key to the nation's energy market, supplying electricity and gas to homes and businesses. The energy company utilizes hydro, solar, wind, and thermal energy sources (Ministry of Business Innovation & Employment [MBIE], 2022). The choice of Genesis Energy for this report stems from its pivotal position in New Zealand's energy industry. At the same time, it deals with its supply chain issues regarding renewable energy transition and fuel supply disturbances (Genesis Energy, 2023). Genesis Energy has implemented digital transformation and technological and retail innovations, enhancing the company's supply chain operational efficiency (Genesis Energy, 2024).

This report will analyse Genesis Energy’s supply chain framework and its main challenges and develop strategies for enhancement and future direction recommendations. The business runs the Huntly Power Station, the largest of this type in the nation, a key electricity generation site. The Huntly power station's contribution to the nation's electricity is approximately 2 percent. Huntly exists as the largest power station in Aotearoa yet will take an essential part in future energy generation as the country transitions to low-carbon systems. Besides making power, Genesis also has services linked to energy, like tools for managing electricity and solar options next to smart home systems(Genesis Energy, 2023).

2. Reason for Choosing Genesis Energy

The main reason to select Genesis Energy is it plays a vital position in New Zealand’s energy market and its complex business operations. Genesis Energy runs a complex supply chain through its widespread energy operations, combining hydropower generation with thermal and solar power generation (Genesis Energy, 2023). The company's push to adopt renewable energy results in supply chain obstacles because of the difficulty of incorporating renewable sources into power frameworks and handle time-dependent energy generation fluctuations

(IEA, 2022). This change allows organizations to study effective strategies that strengthen supply chain resilience. Genesis Energy still relies on fossil fuels, particularly at the Huntly Power Station. Third parties affect supply chain operations through global fuel market vagaries, regulatory demands, and environmental concerns (Genesis Energy, 2023). The company dedicates resources to digital innovation by deploying smart grids and AI-enabled demand prediction systems to optimize its operational capability (Genesis Energy, 2024). These innovative systems enable valuable assessments of present-day supply chain management through technological influences. The report assesses Genesis Energy's supply chain by completing an evaluation and defining essential challenges before establishing strategic improvements for better operational results and environmental sustainability.

3. Industry and Market Position

Genesis Energy is in the utilities area, mainly in electricity generation and retail energy sales. The energy sector in New Zealand uses renewable sources quite a lot. Genesis Energy is listed on the New Zealand Stock Exchange (NZX) with the symbol GNE The New Zealand Government owns a 51% share of the company (to, 2003). Genesis has worked to move to a more sustainable energy situation while keeping a secure supply(Genesis Energy, 2023). During the 2015–2016 financial year Genesis Energy held the position as New Zealand's biggest electrical and natural gas retail enterprise with 26% and 39% market presence. Genesis Energy generated 14% of New Zealand's electricity production during 2015 as it maintained its position as the third largest electricity producer in the country based on MW capacity and GWh generation and revenue(to, 2003). Annual revenues of Genesis Energy exceed NZD 3 billion as part of its sustained financial performance (Genesis Energy, 2024). Genesis Energy stays profitable through fuel supply management and energy price stabilizing strategic moves, operational cost control, and digital transformation investments. The company is unique by utilizing its diverse power generation capacity, secure customer relations, and digital technology investment. The company offers smart home energy solutions, demand response programs, and predictive analytics tools to boost energy efficiency and interact with customers (Genesis Energy, 2024).

Figure Source - Genesis Energy. (2024). *FY24 modern slavery statement*.

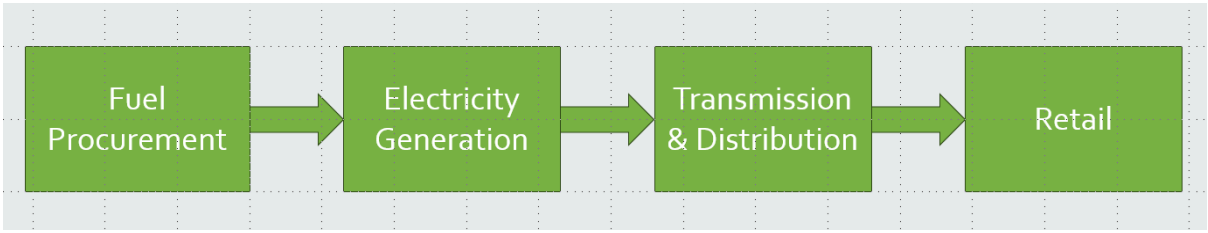
The Map in the figure shows the geographical location of direct suppliers in FY24, including the number of suppliers, the top spend categories, and spending by location.

Genesis Energy is one of the "Big Five" electricity sellers in New Zealand, along with Mercury, Meridian Energy, and Contact Energy, next to Trustpower. In recent times it has given energy to over 500,000 users throughout the country. This means it is the nation's biggest electricity and gas seller (Genesis Energy, 2023). Huntly Power Station runs as a thermal plant which Genesis Energy manages and operates. The plant has two main units that receive power from

coal and gas while also including a 953 MW gas-fired operation. Huntly operates two units from gas and coal combinations while managing a 50 MW open-cycle gas turbine functioning as base load alongside the 403 MW combined cycle gas turbine introduced in June 2007 through an NZ$500 million development. The third coal unit at Huntly exists for emergency purposes and remains ready for use within 90 days. Huntly Power Station's coal units primarily enhance hydroelectric operations during dry winter seasons (to, 2003).

4. Supply Chain Workflow

The supply chain operated by Genesis Energy involves different energy sources spread over multiple infrastructure facilities. The operational sequence of Genesis Energy consists of fuel procurement and energy generation through transmission and distribution and retail operation. The management system combines typical and renewable electric power generation while using digital systems to maximize operational efficiency. The supply chain workflow functions according to the following sequence:



4.1 Fuel Procurement

Genesis Energy sources coal, natural gas, and biomass for thermal energy production at the Huntly Power Station (Genesis Energy, 2023). The Huntly Power Station faces supply chain threats from price instabilities and worldwide factors and environmental governance requirements (International Energy Agency [IEA], 2022). Genesis Energy prescribes to move away from coal toward alternative sustainable energy sources such as torrefied biomass and natural gas which are detailed in the Huntly Power Station firming options report. According to the Genesis FY24 Climate Statement the company aims to decrease its coal-based operations by 2025 by accelerating biomass acquisition and enhancing fuel distribution systems. It has developed a strategic fuel purchasing approach to execute its Gen35 strategy, prioritizing both renewable energy

advancement and system reliability. The company maintains negotiations with storage agreements to procure 10 petajoules of gas, which supports power flexibility during periods of high demand and minimal renewable energy generation (Genesis Energy, 2022). A reliable system for providing biomass supplies stands as the essential foundation for switching Huntly Power Station to renewable energy fuels (Genesis Energy, 2022). The installation of a 100 MW/200 MWh battery storage system at Huntly Power Station moves toward reaching a target 400 MW capacity which will boost grid adaptability and renewable power penetration according to Genesis Energy (2023). The company works through inter-sector partnerships to accelerate the creation of an enhanced renewable-based power system (Genesis Energy, 2022). Genesis Energy uses programs that manage customer energy consumption demands and customer-focused operational improvements to reach its sustainability goals (Genesis Energy, 2023).

4.2 Electricity Generation

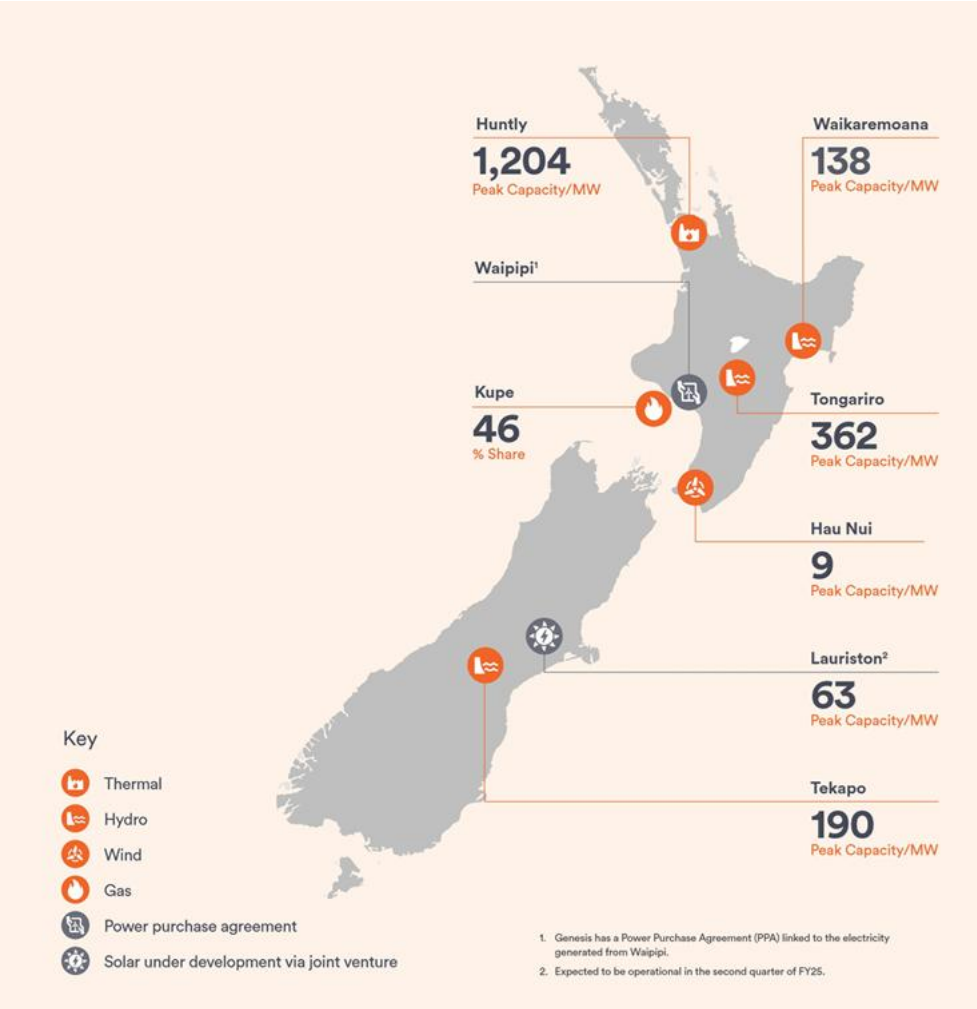


Figure Source - Genesis Energy. (2024). *FY24 modern slavery statement*.

In the above figure mention the types of Power Plants with their electricity generation capacity that is divided into thermal, hydro, wind and gas. All the detailed discussions are in the below sections.

4.2.1 **Thermal Power**: The Huntly Power Station uses coal and gas to create electricity. This aids in keeping the grid stable when demand is high. This plant has the total capacity to generate 1204MW and is located in Huntly, Waikato. Genesis Energy focuses on biomass and natural gas fuel types under New Zealand's carbon emission reduction agreement while ending its coal operation (Genesis Energy, 2023). The company procures its fuel by importing coal and buying gas from local and international suppliers, making its operations susceptible to global market shifts, regulatory changes, and geopolitical factors (IEA, 2022). Domestic natural gas production is subject to fluctuations in availability due to regulatory limits and declining reserves. The shift toward gas as a transition fuel means Genesis must secure long-term supply agreements to maintain energy reliability (Genesis Energy, 2024).

4.2.2 **Hydroelectric Power**: Genesis Energy relies heavily on hydroelectric power as its essential renewable energy source, a significant portion of its electricity production sector. The Tongariro Power Scheme, the Waikaremoana Power Scheme, and Tekapo Power Scheme are operated by the company to produce a major portion of their renewable energy output (Genesis Energy, 2023). Hydropower is New Zealand’s most significant power source since it generates approximately 55% of the country’s electricity output (MBIE, 2022). Through its hydroelectric infrastructure, Genesis Energy guarantees affordable electrical power, which the company produces using renewable sources to decrease dependence on non-renewable fuels.

4.2.2.1 **Tongariro Power Scheme** (361.8 MW), Power stations are Rangipo, Tokaanu, and Mangaio. The facility draws its water from the central

North Island by moving it through connected underground tunnels. The facility offers adaptable power production which adjusts its output according to changing seasonal market conditions (Genesis Energy, 2023).

4.2.2.2 **Waikaremoana Power Scheme** (138 MW), The facility operates in the North Island of New Zealand through three power stations named Kaitawa and Tuai, and Piripaua. The power station produces electricity through a water supply from Lake Waikaremoana. Plays a critical role in grid stability, particularly during peak demand periods (Genesis Energy, 2023).

4.2.2.3 **Tekapo Power Scheme** (190 MW), The power station operates in the South Island while capturing water resources from Lake Tekapo. The production draws from the Waitaki hydro facilities to combine efficient water management techniques. The project works in conjunction with New Zealand's overall renewable energy plan to decrease dependence on thermal power generation (IEA, 2022).

4.2.3 Wind energy: Genesis Energy operates the Hau Nui wind farm in South Wairarapa with an 8.65 MW capacity with 15 wind turbines. The wind energy division of Genesis Energy continues to receive more financial commitment as the company demonstrates its dedication to renewable energy and sustainability. New Zealand relies on wind energy as a key renewable power source that helps decrease carbon pollution and fossil fuel consumption. The wind energy projects at Genesis Energy follow a strategic move towards alternative sustainable power generation methods (Genesis Energy, 2023). Including wind energy within Genesis Energy's supply systems faces complex challenges while creating several growth prospects. The sustainable nature of wind power forces the national grid to implement special measures for managing its variability to maintain steady electricity availability (MBIE, 2022).

4.2.4 Solar energy: The business puts funds into big solar generation and gives solar panel setup to home clients. Genesis Energy is developing solar farms in Manawatu, Waikato, and Hawke's Bay with a combined capacity of 400 MW, aiming to power about 85,000 homes annually. The company Genesis Energy

maintains a continued expansion of its solar power projects to fulfill its renewable energy and sustainability goals. Solar energy remains essential to Genesis Energy because it helps the company decrease its dependence on fossil fuels and establishes a more diverse energy supply mix and resilient supply networks. The implementation of solar power meets New Zealand's national energy targets for 100% renewable electricity generation by 2050 according to MBIE (2022).

- Manawatu: A 200 MW solar farm is in the Manawatu region.
- Waikato: A 100 MW solar farm is in the Waikato region.
- Hawke's Bay: A 100 MW solar farm is in the Hawke's Bay region.

4.2.4.1 Lauriston Solar Farm (Under Development) - The major solar undertaking of Genesis Energy sits within the Canterbury region. This solar facility will produce electricity exceeding 90 GWh annually, which amounts to the power needed for about 15 thousand homes. Genesis Energy uses this project to build its commitment toward carbon emission reduction while strengthening electrical grid stability (Genesis Energy, 2023).

4.2.4.2 Solar Partnerships and Retail Offerings - The energy company Genesis Energy has launched solar power solutions that benefit homeowners as well as businesses. Customers who obtain solar panels from Genesis Energy can sell their unused electricity generation through the Solar Buyback Scheme, which allows them to share the power grid. The company maintains collaborative relations with solar technology businesses to optimize solar panel functioning and smart grid compatibility (Genesis Energy, 2024).

4.2.4.3 Grid-Scale Solar Development - The company aims to build bigger solar power facilities across New Zealand to improve coordination alongside hydro and wind production. The company plans to spend money on battery technologies that stabilize solar power output while guaranteeing energy stability (IEA, 2022).

4.3. Transmission and Distribution:

Genesis Energy Uses Transpower and regional lines companies to Transmit and distribute electricity across New Zealand. The government-owned national grid Transpower spans over 11,000 kilometers and includes 170 substations that help high-voltage electricity reach regional substations. From there, local lines companies handle distribution to end users, and in many places, Genesis Energy also handles to end users. Transmission and distribution costs account for roughly 37.5% of the average energy bill for Genesis customers, covering infrastructure maintenance and operation (Genesis Energy, 2022).

4.4. Retail Operations

Through its retail operations, Genesis Energy transforms energy to serve about 500 thousand Genesis and Frank brand customers who receive electricity and natural gas and LPG services supported by sustainable customer-focused technological developments. Genesis Energy advances its business by implementing a cost-effective retail process model and digital CRM tools to improve customer interaction while integrating renewable energy solutions under the Gen35 framework. The company operates sustainability-focused initiatives to match business goals and market position (Genesis Energy, 2022; Hyde et al., n.d.). Genesis installed smart metering and home energy management tools that help customers monitor and manage their energy usage in real-time. These types of technological advancements help companies grow and sustain themselves.

5. Supply Chain Challenges

Genesis Energy has various challenges that affect its consistent energy generation and delivery. An ongoing gas shortage condition has forced Genesis Energy to build up coal reserves at Huntly Power Station to more than 500 kilotonnes. The procurement of coal from Indonesia becomes more complex because the country implements export quotas and imposes regulatory restrictions on importation. The energy supply in New Zealand has become inconsistent because it relies too much on renewable energy sources. The combination of wind power interruptions and decreased hydro lake water levels compels

Genesis to operate its thermal generators for backup purposes. Fuel supply and storage capacity must be diversified because the country depends on them. Genesis Energy depends on non-renewable power sources for backup whenever electricity fluctuates in renewable resources like low water because of low rainfall, low windy air, and low sunlight in the solar panel. Coal dependence restricts Genesis Energy's sustainability. During the import of coal, there are also many problems like delays in port due to conjunction, which affect companies' commitment (Genesis Energy, 2024; Genesis Energy, 2025; Hyde et al., n.d.).

6. Analysis of the Solution

Genesis Energy can tackle its supply chain problems with a plan that tries to find a middle ground between current demand and lasting sustainability goals. To fix the gas shortage, Genesis boosted coal amounts at Huntly Power Station - the company has 500 kilotonnes set aside to help make energy at times of greatest need, but they can also switch to permanent solutions by doing research and development in biogas and geothermal.

As we know, New Zealand is known as the "land of geothermal" due to its abundant geothermal resources, particularly in the central North Island's Taupō Volcanic Zone; I think in my point of view, we have to use that energy flow as much as possible. The solution of adding stocks of coal gives them stable operations, but on the other hand, it also clashes with their Gen35 pledge to reduce carbon emissions long-term. Getting coal from Indonesia makes issues more complex. Regulatory rules besides export limits add logistical and geopolitical dangers. Genesis seeks ways to lessen coming disturbances and also arranges possibilities for storing gas, a maximum of 10 petajoules(pj). This project betters robustness because it gives additional ability plus adaptability when supply is shaky. Aside from the steps at hand, Genesis progresses greatly as it shifts to the energy gained from replenishment. Capital spent on biomass fuel, destined for Huntly Power Station, aims to dismiss coal reliance - energy work will match sustainability aims. To manage the changeability of wind besides hydro hold, adding battery storage assists with better consolidating these sources.

Genesis Energy also tries its best towards sustainability by executing investments through investments in Lauriston Solar Farm together with the Tongariro Hydroelectric Scheme (Genesis Energy, 2023). Efficient supply chain coordination stands essential for achieving the

integration of renewable energy into current electric grid systems (MBIE, 2022). The company's key strategy includes developing retail and technology solutions for advanced distribution and demand management systems (Genesis Energy, 2024). The ups and downs of renewable energy production from solar and wind create supply chain problems for keeping electricity flow consistent. AI can also help them in both ways for the customer experience while enrolling to know their billing cycle and to predict coming demand more precisely. They can also apply capacity planning models to assess future energy demands and determine the required infrastructure investments.

7. Recommendations

The company should establish multi-year biomass supplier contracts for continuous fuel delivery. The organization should increase investments for grid modernization to boost renewable energy efficiency. The application of demand forecasting models enables teams to optimize production and the distribution of energy supplies. The company should establish stronger alliances with research organizations to develop innovative energy solutions. They should use digital transformation methods to make their supply chains more resilient alongside improving customer demand predictions.

8. Conclusion

The supply chain at Genesis Energy is in the middle of a major transformation because the company is transitioning to renewable energy while rebuilding its power grid networks and modernizing digital information flows. The company faces fuel procurement problems, transmission limitations, and supply chain interruptions while its technological and sustainability strategy fosters long-term business strength. Genesis Energy will deliver sustainable energy and national and global sustainability targets through innovative solutions and effective risk management procedures.

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